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GATE ECE · ENGINEERING MATHEMATICS · CHAPTER-WISE PYQS

Linear Algebra

GATE ECE Engineering Mathematics · Previous Year Questions with Answer Key

54	30	1994 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	38 questions
Marks 2	16 questions
Answer key	at the end

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Marks 1 38 questions

Q1. GATE ECE 2026

Consider the matrix $M = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 0 \\ -1 & a & b \end{bmatrix}$.

Which of the following options is/ are TRUE if $\det(M) \neq 0$?

(A) $a = -\frac{1}{2}$ and $b = -\frac{1}{2}$

(B) $a = \frac{1}{2}$ and $b = \frac{1}{2}$

(C) $a = -3$ and $b = 0$

(D) $a = \frac{1}{2}$ and $b = -3$

Open solution

Q2. GATE ECE 2025

Consider the matrix A below:

$$A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & \alpha & \beta \\ 0 & 0 & 0 & \gamma \end{bmatrix}$$

For which of the following combinations of α, β and γ, is the rank of A at least three?

(i) $\alpha = 0$ and $\beta = \gamma \neq 0$

(ii) $\alpha = \beta = \gamma = 0$

(iii) $\beta = \gamma = 0$ and $\alpha \neq 0$

(iv) $\alpha = \beta = \gamma \neq 0$

(A) Only (i), (iii), and (iv)

(B) Only (iv)

(C) Only (ii)

(D) Only (i) and (iii)

Open solution

Q3. GATE ECE 2024

Let $\mathbb{R}$ and $\mathbb{R}^3$ denote the set of real numbers and the three dimensional vector space over it, respectively. The value of α for which the set of vectors

$\{[2 \quad -3 \quad \alpha], [3 \quad -1 \quad 3], [1 \quad -5 \quad 7]\}$

does not form a basis of $\mathbb{R}^3$ is _____.

Open solution

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Q4. GATE ECE 2023

Let the sets of eigenvalues and eigenvectors of a matrix B be $\{\lambda_k | 1 \leq k \leq n\}$ and $\{v_k | 1 \leq k \leq n\}$, respectively. For any invertible matrix P, the sets of eigenvalues and eigenvectors of the matrix A, where $B = P^{-1}AP$, respectively, are

Open solution

(A) $\{\lambda_k \det(A) | 1 \leq k \leq n\}$ and $\{Pv_k | 1 \leq k \leq n\}$

(B) $\{\lambda_k | 1 \leq k \leq n\}$ and $\{v_k | 1 \leq k \leq n\}$

(C) $\{\lambda_k | 1 \leq k \leq n\}$ and $\{Pv_k | 1 \leq k \leq n\}$

(D) $\{\lambda_k | 1 \leq k \leq n\}$ and $\{P^{-1}v_k | 1 \leq k \leq n\}$

Q5. GATE ECE 2022

Consider a system of linear equations Ax = b, where

$$A = \begin{bmatrix} 1 & -\sqrt{2} & 3 \\ -1 & \sqrt{2} & -3 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Open solution

This system is equations admits _____.

(A) a unique solution for x

(B) infinitely many solutions for x

(C) no solutions for x

(D) exactly two solutions for x

Q6. GATE ECE 2021

If the vectors (1.0, −1.0, 2.0), (7.0, 3.0, x) and (2.0, 3.0, 1.0) in R^3 are linearly dependent, the value of x is _____.

Open solution

Q7. GATE ECE 2020

If $\mathbf{v_1}, \mathbf{v_2} \ldots \mathbf{v_6}$ are six vectors in $\mathbb{R}^4$, which one of the statements is FALSE?

Open solution

(A) Any four of these vectors form a basis for $\mathbb{R}^4$.

(B) It is not necessary that these vectors span $\mathbb{R}^4$.

(C) If $\{\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5, \mathbf{v}_6\}$ spans $\mathbb{R}^4$, then it forms a basis of $\mathbb{R}^4$.

(D) These vectors are not linearly independent.

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Q8. GATE ECE 2018

Let **M** be a real 4×4 matrix. Consider the following statements :

S1: **M** has 4 linearly independent eigenvectors.

S2: **M** has 4 distinct eigenvalues.

S3: **M** is non-singular (invertible).

Which one among the following is TRUE?

(A) S1 implies S2 (B) S2 implies S1 (C) S1 implies S3 (D) S3 implies S2

Open solution

Q9. GATE ECE 2018

Consider matrix $A = \begin{bmatrix} k & 2k \\ k^2 - k & k^2 \end{bmatrix}$ and

vector $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

The number of distinct real values of k for which the equation AX = 0 has infinitely many solutions is _____.

Open solution

Q10. GATE ECE 2017 Set 1

The rank of the matrix $M = \begin{bmatrix} 5 & 10 & 10 \\ 1 & 0 & 2 \\ 3 & 6 & 6 \end{bmatrix}$ is

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q11. GATE ECE 2017 Set 1

Consider the 5×5 matrix $A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \\ 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 1 & 2 \\ 2 & 3 & 4 & 5 & 1 \end{bmatrix}$

It is given that A has only one real eigen value. Then the real eigen value of A is

(A) -2.5 (B) 0 (C) 15 (D) 25

Open solution

Q12. GATE ECE 2016 Set 2

The value of x for which the matrix $A = \begin{bmatrix} 3 & 2 & 4 \\ 9 & 7 & 13 \\ -6 & -4 & -9 + x \end{bmatrix}$ has zero as an eigen value is _____.

Open solution

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Q13. GATE ECE 2016 Set 1

Let $M^4 = \mathrm{I}$ (where I denotes the identity matrix) and $M \neq \mathrm{I}$, $M^2 \neq \mathrm{I}$ and $M^3 \neq \mathrm{I}$. Then, for any natural number k, M^{-1} equals:

(A) M^{4k+1} (B) M^{4k+2} (C) M^{4k+3} (D) M^{4k}

Open solution

Q14. GATE ECE 2016 Set 3

Consider a 2×2 square matrix $A = \begin{bmatrix} \sigma & x \\ \omega & \sigma \end{bmatrix}$

Where x is unknown. If the eigenvalues of the matrix A are $(\sigma + j\omega)$ and $(\sigma - j\omega)$, then x is equal to

(A) $+j\omega$ (B) $-j\omega$ (C) $+\omega$ (D) $-\omega$

Open solution

Q15. GATE ECE 2015 Set 3

For $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, the determinant of $A^T A^{-1}$ is

(A) $\sec^2 x$ (B) $\cos 4x$ (C) 1 (D) 0

Open solution

Q16. GATE ECE 2015 Set 2

The value of 'x' for which all the eigenvalues of the matrix given below are real is $\begin{bmatrix} 10 & 5+j & 4 \\ x & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix}$

(A) $5 + j$ (B) $5 - j$ (C) $1 - 5j$ (D) $1 + 5j$

Open solution

Q17. GATE ECE 2015 Set 1

Consider system of linear equations :

$$x - 2y + 3z = -1$$

$$x - 3y + 4z = 1$$

and

$$-2x + 4y - 6z = k,$$

The value of 'k' for which the system has infinitely many solutions is _____.

Open solution

Q18. GATE ECE 2015 Set 1

The value of 'P' such that the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is an eigenvector of the matrix $\begin{bmatrix} 4 & 1 & 2 \\ P & 2 & 1 \\ 14 & -4 & 10 \end{bmatrix}$ is _____.

Open solution

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Q19. GATE ECE 2014 Set 1

For matrices of same dimension M, N and scalar c, which one of these properties **DOES NOT ALWAYS hold** ?

(A) $\left(M^T\right)^T = M$

(B) $(cM)^T = c(M)^T$

(C) $(M+N)^T = M^T + N^T$

(D) $MN = NM$

Open solution

Q20. GATE ECE 2014 Set 1

A real (4×4) matrix A satisfies the equation $A^2 = \mathrm{I}$, where I is the (4×4) identity matrix. The positive eigen value of A is _____.

Open solution

Q21. GATE ECE 2014 Set 1

Consider the matrix $J_6 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Which is obtained by reversing the order of the columns of the identity matrix I_6. Let $P = \mathrm{I}_6 + \alpha\ J_6$, where α is a non $-$ negative real number. The value of α for which det $(P) = 0$ is _____.

Open solution

Q22. GATE ECE 2014 Set 2

The determinant of matrix A is 5 and the determinant of matrix B is 40. The determinant of matrix AB is _____.

Open solution

Q23. GATE ECE 2014 Set 2

The system of linear equations $\begin{pmatrix} 2 & 1 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 5 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 14 \end{pmatrix}$ has

(A) a unique solution

(B) infinitely many solutions

(C) no solution

(D) exactly two solutions

Open solution

Q24. GATE ECE 2014 Set 2

The maximum value of the determinant among all 2×2 real symmetric matrices with trace 14 is _____.

Open solution

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Q25. GATE ECE 2014 Set 3

Which one of the following statements is NOT true for a square matrix A?

(A) If A is upper triangular, the eigenvalues of A are the diagonal elements of it

(B) If A is real symmetric, the eigenvalues of A are always real and positive

(C) If A is real , the eigenvalues of A and A^T are always the same

(D) If all the principal minors of A are positive , all the eigenvalues of A are also positive

Open solution

Q26. GATE ECE 2013

The minimum eigenvalue of the following matrix is $\begin{bmatrix} 3 & 5 & 2 \\ 5 & 12 & 7 \\ 2 & 7 & 5 \end{bmatrix}$

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q27. GATE ECE 2013

Let A be an $m \times n$ matrix and B an $n \times m$ matrix. It is given that determinant $(\mathrm{I}_m + AB) =$determinant $(\mathrm{I}_n + BA)$, where I_k is the $k \times k$ identity matrix. Using the above property, the determinant of the matrix given below is $\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$

(A) 2 (B) 5 (C) 8 (D) 16

Open solution

Q28. GATE ECE 2012

Given that $A = \begin{bmatrix} -5 & -3 \\ 2 & 0 \end{bmatrix}$ and $\mathrm{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the value of A^3 is

(A) $15A + 12\,\mathrm{I}$ (B) $19A + 30\,\mathrm{I}$ (C) $17A + 15\,\mathrm{I}$ (D) $17A + 21\,\mathrm{I}$

Open solution

Q29. GATE ECE 2011

The system of equations $x + y + z = 6$, $x + 4y + 6z = 20$, $x + 4y + \lambda z = \mu$ has no solution for values of λ and μ given by

(A) $\lambda = 6,\ \mu = 20$ (B) $\lambda = 6,\ \mu \neq 20$ (C) $\lambda \neq 6,\ \mu = 20$ (D) $\lambda \neq 6,\ \mu \neq 20$

Open solution

Q30. GATE ECE 2010

The eigen values of a skew-symmetric matrix are

(A) always zero (B) always pure imaginary

(C) either zero (or) pure imaginary (D) always real

Open solution

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Q31. GATE ECE 2008

All the four entries of $2\ x\ 2$ matrix $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$ are non-zero and one of the eigen values is zero. Which of the following statement is true?

Open solution

(A) $P_{11}\,P_{22} - P_{12}\,P_{21} = 1$
(B) $P_{11}\,P_{22} - P_{12}\,P_{21} = -1$
(C) $P_{11}\,P_{22} - P_{21}\,P_{12} = 0$
(D) $P_{11}\,P_{22} + P_{12}\,P_{21} = 0$

Q32. GATE ECE 2008

The system of linear equations $\left.\begin{matrix} 4x + 2y = 7 \\ 2x + y = 6 \end{matrix}\right\}$ has

Open solution

(A) a unique solution
(B) no solution
(C) an infinite no. of solutions
(D) exactly two distinct solution.

Q33. GATE ECE 2006

For the matrix $\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$. The eigen value corresponding to the eigen vector $\begin{bmatrix} 101 \\ 101 \end{bmatrix}$ is

Open solution

(A) 2
(B) 4
(C) 6
(D) 8

Q34. GATE ECE 2006

The rank of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ is

Open solution

(A) 0
(B) 1
(C) 2
(D) 3

Q35. GATE ECE 2000

The eigen values of the matrix $\begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & -1 & 4 \end{bmatrix}$ are

Open solution

(A) $2, -2, 1, -1$
(B) $2, 3, -2, 4$
(C) $2, 3, 1, 4$
(D) None

Q36. GATE ECE 1998

The eigen values of the matrix $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ are

Open solution

(A) $1, 1$
(B) $-1, -1$
(C) $j, -j$
(D) $1, -1$

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Q37. GATE ECE 1994

The rank of $(m \times n)$ matrix $(m < n)$ cannot be more then

(A) m (B) n (C) mn (D) None

Open solution

Q38. GATE ECE 1994

The following system of equations

$x_1 + x_2 + x_3 = 3$

$x_1 - x_3 = 0$

$x_1 - x_2 + x_3 = 1$ has

(A) Unique solution

(B) No solution

(C) Infinite number of solutions

(D) Only one solution

Open solution

Marks 2 16 questions

Q39. GATE ECE 2025

Consider the vectors

$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, b = \begin{bmatrix} 0 \\ 3\sqrt{2} \end{bmatrix}$

For real-valued scalar variable x, the value of

$\min_x \|ax - b\|_2$

is_____(rounded off to two decimal places).

$\|\cdot\|_2$ denotes the Euclidean norm, i.e., for $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \|y\|_2 = \sqrt{y_1^2 + y_2^2}$.

Open solution

Q40. GATE ECE 2024

Consider the matrix $\begin{bmatrix} 1 & k \\ 2 & 1 \end{bmatrix}$, where k is a positive real number. Which of the following vectors is/are eigenvector(s) of this matrix?

(A) $\begin{bmatrix} 1 \\ -\sqrt{2/k} \end{bmatrix}$

(B) $\begin{bmatrix} 1 \\ \sqrt{2/k} \end{bmatrix}$

(C) $\begin{bmatrix} \sqrt{2k} \\ 1 \end{bmatrix}$

(D) $\begin{bmatrix} \sqrt{2k} \\ -1 \end{bmatrix}$

Open solution

Q41. GATE ECE 2023

Let x be an $n \times 1$ real column vector with length $l = \sqrt{x^T x}$. The trace of the matrix $P = xx^T$ is

(A) l^2 (B) $\frac{l^2}{4}$ (C) l (D) $\frac{l^2}{2}$

Open solution

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Q42. GATE ECE 2023

Open solution

The state equation of a second order system is

$\dot{x}(t) = Ax(t), \quad x(0)$ is the initial condition.

Suppose λ_1 and λ_2 are two distinct eigenvalues of A and v_1 and v_2 are the corresponding eigenvectors. For constants α_1 and α_2, the solution, $x(t)$, of the state equation is

(A) $\sum_{i=1}^{2} \alpha_i e^{\lambda_i t} v_i$

(B) $\sum_{i=1}^{2} \alpha_i e^{2\lambda_i t} v_i$

(C) $\sum_{i=1}^{2} \alpha_i e^{3\lambda_i t} v_i$

(D) $\sum_{i=1}^{2} \alpha_i e^{4\lambda_i t} v_i$

Q43. GATE ECE 2022

Open solution

Let α, β two non-zero real numbers and v_1, v_2 be two non-zero real vectors of size 3×1. Suppose that v_1 and v_2 satisfy $v_1^T v_2 = 0$, $v_1^T v_1 = 1$ and $v_2^T v_2 = 1$. Let A be the 3×3 matrix given by :

A = $\alpha v_1 v_1^T + \beta v_2 v_2^T$

The eigen values of A are _____.

(A) 0, α, β

(B) 0, $\alpha + \beta$, $\alpha - \beta$

(C) 0, $\frac{\alpha+\beta}{2}$, $\sqrt{\alpha\beta}$

(D) 0, 0, $\sqrt{\alpha^2 + \beta^2}$

Q44. GATE ECE 2021

Open solution

A real 2×2 non-singular matrix A with repeated eigen value is given as

$$A = \begin{bmatrix} x & -3.0 \\ 3.0 & 4.0 \end{bmatrix}$$

where x is a real positive number. The value of x (rounded off to one decimal) is _____

Q45. GATE ECE 2020

Open solution

Consider the following system of linear equations.

$x_1 + 2x_2 = b_1; 2x_1 + 4x_2 = b_2; 3x_1 + 7x_2 = b_3; 3x_1 + 9x_2 = b_4$

Which one of the following conditions ensures that a solution exists for the above system?

(A) $b_2 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

(B) $b_3 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$

(C) $b_2 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$

(D) $b_3 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

Q46. GATE ECE 2017 Set 2

Open solution

The rank of the matrix $\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$ is _____.

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Q47. GATE ECE 2016 Set 2

The matrix $A = \begin{bmatrix} a & 0 & 3 & 7 \\ 2 & 5 & 1 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & b \end{bmatrix}$ has det $(A) = 100$ and trace $(A) = 14$. The value of $|a - b|$ is _____.

Open solution

Q48. GATE ECE 2016 Set 1

A sequence $x[n]$ is specified as

$$\begin{bmatrix} x[n] \\ x[n-1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \ for\ n \geq 2.$$

The initial conditions are $x[0] = 1,\ x[1] = 1$ and $x[n] = 0$ for $n < 0$. The value of $x[12]$ is _____.

Open solution

Q49. GATE ECE 2016 Set 3

If the vectors $e_1 = (1, 0, 2)$, $e_2 = (0, 1, 0)$ and $e_3 = (-2, 0, 1)$ form an orthogonal basis of the three dimensional real space R^3, then the vectors $u = (4, 3, -3) \in R^3$ can be expressed as

(A) $u = -\frac{2}{5}e_1 - 3e_2 - \frac{11}{5}e_3$

(B) $u = -\frac{2}{5}e_1 - 3e_2 + \frac{11}{5}e_3$

(C) $u = -\frac{2}{5}e_1 + 3e_2 + \frac{11}{5}e_3$

(D) $u = -\frac{2}{5}e_1 + 3e_2 - \frac{11}{5}e_3$

Open solution

Q50. GATE ECE 2009

The eigen values of the following matrix $\begin{bmatrix} -1 & 3 & 5 \\ -3 & -1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$ are

(A) $3, 3 + 5j, 6 - j$

(B) $-6 + 5j, 3 + j, 3 - j$

(C) $3 + j, 3 - j, 5 + j$

(D) $3, -1 + 3j, -1 - 3j$

Open solution

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Q51. GATE ECE 2006

The eigen values and the correspondinng eigen vectors of a 2×2 matrix are given by

Eigen value

$\lambda_1 = 8$

$\lambda_2 = 4$

Eigen vector

$V_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$V_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is

(A) $\begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}$

(B) $\begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$

(D) $\begin{bmatrix} 4 & 8 \\ 8 & 4 \end{bmatrix}$

Open solution

Q52. GATE ECE 2005

Given an orthogonal matrix $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$ then the value of $\left(AA^T\right)^{-1}$ is

(A) $\frac{1}{4}\mathrm{I}_4$

(B) $\frac{1}{2}\mathrm{I}_4$

(C) I

(D) $\frac{1}{3}\mathrm{I}_4$

Open solution

Q53. GATE ECE 2005

Given the matrix $\begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix}$, the eigen vector is

(A) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

(B) $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$

(C) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$

(D) $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

Open solution

Q54. GATE ECE 2005

If $A = \begin{bmatrix} 2 & -0.1 \\ 0 & 3 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1/2 & a \\ 0 & b \end{bmatrix}$ then $a + b =$

(A) $\frac{7}{20}$

(B) $\frac{3}{20}$

(C) $\frac{19}{60}$

(D) $\frac{11}{20}$

Open solution

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Answer Key

Marks 1

1. **B, D**	2. **A**	3. **5**	4. **C**	5. **C**
6. **8**	7. **A**	8. **B**	9. **2 to 2**	10. **C**
11. **C**	12. **1**	13. **C**	14. **D**	15. **C**
16. **B**	17. **2**	18. **17**	19. **D**	20. **1**
21. **1**	22. **200**	23. **B**	24. **49**	25. **B, D**
26. **A**	27. **B**	28. **B**	29. **B**	30. **C**
31. **C**	32. **B**	33. **C**	34. **C**	35. **B**
36. **D**	37. **A**	38. **A**		

Marks 2

39. **3**	40. **A, B**	41. **A**	42. **A**	43. **A**
44. **10**	45. **C**	46. **4**	47. **3**	48. **233**
49. **D**	50. **D**	51. **A**	52. **C**	53. **D**
54. **A**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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